



Toward open science in PLS-SEM: Assessing the state of the art and future perspectives

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ABSTRACT

Driven by the high-profile failures to reproduce and replicate published findings, there have been increasing demands to adopt open science practices across scientific disciplines in order to enhance research transparency. Critics have highlighted the use of underpowered studies and researchers' analytical degrees of freedom as factors contributing to these issues. Despite methodological advances and updated guidelines, similar concerns persist regarding studies utilizing partial least squares structural equation modeling (PLS-SEM). Open science practices can help alleviate these concerns by facilitating transparency in PLS-SEM-based studies. However, the current level of adherence to these practices remains unknown. In this article, we conduct a comprehensive literature review of leading marketing journals to assess the extent to which open science practices are implemented in PLS-SEM-based studies. Based on the observed lack of adoption, we propose a PLS-SEM-specific preregistration template that researchers can use to foster transparency in their analyses, thereby bolstering confidence in their findings.

1. Introduction

In recent decades, scientific circles have faced a crisis of confidence as a result of several high-profile failures in reproducing and replicating published findings (Miller & Ulrich, 2022; Open Science Collaboration, 2015; Pashler & Wagenmakers, 2012). Researchers have identified several key factors contributing to these anomalies, including underpowered original studies with small sample sizes, authors' analytical degrees of freedom, and the pressure to publish "successful" research with novel and interesting findings based on statistically significant results (Bakker et al., 2012; Etz & Vandekerckhove, 2016; Ioannidis, 2005; Simmons et al., 2011; Wicherts et al., 2016). In response, the scientific community has increasingly called for the adoption of open science practices, such as preregistration (i.e., specifying the research plan in advance), registered reports, and the provision of access to research materials, to enhance research transparency (de Roure et al., 2010;

Grand et al., 2012; Nosek et al., 2018; Vicente-Saez & Martinez-Fuentes, 2018).

The dual issues of underpowered studies with small sample sizes and researchers' analytical degrees of freedom have also plagued empirical research conducted via partial least squares structural equation modeling (PLS-SEM), a component-based technique that offers considerable strengths in estimating complex relationships between constructs and their indicators. For example, PLS-SEM has been misapplied in potentially underpowered studies due to its ability to converge to a solution even with small samples (Hair et al., 2019; Marcoulides & Saunders, 2006). Researchers also have the flexibility to choose from among a plethora of analytical options, such as different kinds of software with varying settings, structural and measurement model metrics, bootstrap methods and samples, modes of measurement, higher-order constructs, missing data treatments, and moderation methods, among others. In any given PLS-SEM application, researchers often explore

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multiple versions of structural or measurement models in the background (Petter & Hadavi, 2021). All of these analytical degrees of freedom, or a combination of them, could lead to differences in results in the same dataset (Becker et al., 2023).¹ However, these analytical choices are frequently undisclosed in published studies due to a variety of reasons, including publication bias and limited journal space (Petter & Hadavi, 2021). This lack of transparency impedes the independent verification of published research and allows false positive findings to persist (Nosek et al., 2012).

Recent research in PLS-SEM has provided updated methodological guidelines to promote robust application and better reporting of results (Cheah et al., 2019; Hair et al., 2019; Marcoulides & Chin, 2013; Sarstedt, Hair, Pick, et al., 2022a). Despite these developments, an important gap in the literature pertains to the extent of the adoption of open science practices in PLS-SEM applications and how they can be effectively implemented to enhance transparency.

The inability to independently verify the reproducibility and replicability of results poses a risk to the confidence placed in published research, including PLS-SEM-based studies. This concern is magnified in light of recent critical assessments of PLS-SEM applications (Schuberth et al., 2021). Open science practices offer a way forward to bolster the analytical rigor of PLS-SEM studies by promoting transparency regarding researchers' analytical choices and providing them with a platform to host and freely share research materials on repositories such as the Open Science Framework (OSF; Nosek et al., 2012).²

Our article is situated at the intersection of ongoing discussions surrounding the need for open science in general and the robust application of PLS-SEM methodology in particular. It contributes to the field in the following key ways: First, we conduct a systematic literature review of articles published in leading marketing journals (SCImago Q1, SCImago, n.d.). The aim is to assess the degree to which open science practices are adhered to in applied PLS-SEM research. Through this analysis, we aim to document the degree to which PLS-based studies have embraced transparency and facilitated the independent verification of published results regarding reproducibility and replicability. Notably, despite more than a decade of advocacy for open science, we find persistently low levels of adoption of open science practices (e.g., preregistration; open materials, data, code, or model files), as well as limited replication efforts in PLS-SEM-based studies published in marketing journals. This status quo does not align with the ideal conduct of error-correcting science and may perpetuate false positive findings.

Second, given the observed lack of adherence to open science practices, we advocate the open sharing of resources whenever possible (e.g., study materials, data, and model files). In addition, we propose a preregistration template specifically tailored to PLS-SEM applications. Researchers can use this template to support the independent reproducibility and replicability of their findings, thereby bolstering

confidence in their studies. Such a template is useful because PLS-SEM is often used in exploratory research for complex modeling tasks that involve multiple subjective yet justifiable analytical decisions. These decisions can potentially influence the statistical significance of results and pave the way for publication bias (Hair et al., 2019). By adopting open science practices and utilizing the suggested preregistration template, PLS-SEM applications can establish a transparent and trust-enhancing mechanism. This approach offers powerful assurances that empirical analyses were conducted as stated and in accordance with the latest guidelines.

2. Background

2.1. Reproducibility and replicability in research

Reproducibility and replicability are two fundamental characteristics that ensure the error-correcting nature of science (Merton, 1973). Reproducibility refers to the ability to independently verify statistical results obtained from a given dataset by using the same dataset. By contrast, replicability involves independently verifying conclusions about a phenomenon of interest (e.g., hypothesized relationship between two or more variables) across multiple studies with data collected from the same population (Köhler & Cortina, 2021).³ Thus, reproducibility focuses on verifying specific results, which requires access to the original study's artifacts, including code, data, and details of the applied methodology. Conversely, replicability centers on the research question and requires detailed information about the original study's research design, sampling, and methodology, so that others may recreate artifacts that closely resemble the original study to test the original hypotheses (National Academies of Sciences, 2020, chapter 3). In practice, replication studies can vary in their proximity to the original study. They range from direct replications, which closely resemble the original study, to conceptual replications that involve some deviation (Brandt et al., 2014).

In contrast to the physical and natural sciences, the field of management sciences has historically put less emphasis on reproducibility and replicability, instead focusing on the novelty and generalizability of findings.⁴ Top journals regularly require novel and statistically significant findings and discourage the publication of studies that prioritize reproducibility and replicability (Nosek et al., 2012; Tipu & Ryan, 2022). This emphasis can inadvertently lead researchers to engage in questionable research practices, such as selective reporting, *p*-hacking, and "hypothesizing after the results are known" (HARKing; John et al., 2012; Kepes et al., 2022; Miller & Ulrich, 2022).⁵

These factors have contributed to publication bias and sparked debates about the reproducibility and replicability of scientific findings (e.g., Ioannidis, 2005). Over the past decade, the psychological sciences, in particular, have witnessed some of the most intense debates among scholars regarding the trustworthiness and reliability of findings

¹ It is important to recognize that the issue of researchers' degree of freedom is not exclusive to PLS-SEM but affects all statistical methods (Simmons et al., 2011; Wicherts et al., 2016). For example, Silberzahn et al. (2018) asked 29 research teams to use the same dataset to answer the same question using regression techniques. The results revealed substantial variation in the conclusions drawn, stemming from justifiable yet subjective analytic choices made by the teams. The authors note (p. 338): "These findings suggest that significant variation in the results of analyses of complex data may be difficult to avoid, even by experts with honest intentions." To address this, they advocate for transparency in reporting to enable others to retrace the analysis paths. Users of covariance-based SEM face similar choices, especially when seeking to achieve model identification by constraining or freeing paths or by adding constructs (Petter, 2018). Thus, our focus on PLS-SEM should not be interpreted as a criticism of the method itself but rather as a discussion regarding the potential for ambiguity in applying the method, coupled with a lack of transparency and the incentives for researchers to discover statistically significant results, which may contribute to false positives.

² <https://osf.io>.

³ There is significant variation in how researchers across different disciplines understand and communicate the terms reproducibility and replicability, which often leads to conflation between the two.

⁴ In contrast to reproducibility and replicability, the concept of generalizability focuses on the consistency of findings across *different* populations. It typically involves the consideration of substantive, theoretically motivated moderators, such as respondents' demographics, characteristics of technologies, and variations in cultural or socio-technical contexts and time periods (Köhler & Cortina, 2021).

⁵ *p*-hacking refers to the practice of selectively reporting those analyses that yield statistically significant results while omitting the numerous statistical analyses and/or data eligibility specifications that do not produce statistically significant results in the background (Simmons et al., 2011). HARKing refers to the presentation of *post hoc* hypotheses as if they were formulated prior to data collection, thus giving the false impression of being *a priori* hypotheses (Kerr, 1998).

(Baumeister et al., 2022; Nosek et al., 2022; Pashler & Wagenmakers, 2012). For example, an influential study by the Open Science Collaboration (2015; OSC) revealed that only 39% of previously published findings in three leading psychology journals were replicable. However, Gilbert et al. (2016) pointed to several limitations and argued that OSC's conclusion was not warranted and that replicability in psychological sciences is actually quite high. The "Many Labs" study by Klein et al. (2014) successfully replicated 10 out of 13 established effects and suggested that replicability depends more on the specific effect than the sample or setting used. Similar efforts to assess the reproducibility of published psychology research based on original data have yielded mixed results (Artner et al., 2021; Hardwicke et al., 2021; Maassen et al., 2020).

Despite these disagreements, scholars unanimously agree there is a critical lack of independent studies that focus on the reproducibility and replicability of published research across all management disciplines (Köhler & Cortina, 2021; Makel et al., 2012). For example, Ryan and Tipu (2022) analyzed articles published in 121 leading management journals from 2008 to 2017 and found that only 1.47% of published research included elements of overt replication. This finding is unsurprising, given the historical editorial policies of leading management journals that have discouraged replications (Nosek et al., 2012; Tipu & Ryan, 2022). However, recent replication efforts within the management disciplines, including marketing (Babin et al., 2021; Lynch et al., 2015) and information systems (Dennis et al., 2020), have sought to challenge the status quo by critically examining their body of research, highlighting the need for improved transparency.

2.2. Open science practices for enhanced transparency and scientific rigor

Amid this ongoing debate, open science practices have been proposed as tools to assess and enhance the reproducibility and replicability of published research (Nosek et al., 2018). Open science is broadly defined as "transparent and accessible knowledge that is shared and developed through collaborative networks" (Vicente-Saez & Martinez-Fuentes, 2018, p. 434). The main goal of these practices is to enable the independent reproduction and replication of published findings and to improve on them (Nelson et al., 2018; Nosek et al., 2012). Open science seeks to enhance research transparency by encouraging practices such as preregistration, registered reports, and open access to study materials, data, and code, whenever feasible and appropriate (Nosek et al., 2018).

More specifically, preregistration means clearly communicating research questions, hypotheses, and study procedures before primary data collection takes place (Nosek et al., 2018). In the case of studies involving existing or secondary data, preregistration can be conducted before data analysis. The main purpose of preregistration is to separate exploratory (hypothesis-generating) research from confirmatory (hypothesis-testing) research (Hardwicke & Wagenmakers, 2023; Nosek et al., 2018).⁶ Exploratory research involves model respecifications and comparisons that inform hypothesis generation, rather than confirming a predefined, theory-derived model. For example, researchers may initially estimate a model guided by nascent theory or logic but subsequently respecify parts of it after data analysis (e.g., by adding relationships to enhance model fit or identifying previously unknown subgroups) to develop an improved theoretical model (Chin, 1998). While this process lays the foundation for subsequent hypothesis-testing studies, it also overfits the resulting model to the dataset used to generate it. For this reason, the dataset used to generate a hypothesis should not be used to test it. Hence, researchers need to differentiate between testing hypotheses derived *a priori* from existing literature and conducting post hoc data analysis to identify relationships that were not hypothesized in advance. Platforms such as the OSF and [aspredicted.org](https://www.cos.io/initiatives/prereg)

create a fixed and time-stamped preregistration document that serves to demonstrate that researchers have not adapted their hypotheses after viewing the results, thereby enhancing the study's trustworthiness by disentangling planned and post hoc analyses.

While authors can conduct preregistrations independently of the peer review process, a registered report—a multi-stage publication format—requires peer review of the study's theoretical foundation, hypotheses, procedure, and analysis plan before primary data collection (or, in the case of secondary data, before analysis).⁷ Articles that successfully pass this peer-review process receive in-principle acceptance regardless of the results obtained, while still allowing the flexibility to report serendipitous findings (Nosek & Lakens, 2014). Literature comparing registered reports and standard reports suggests that registered reports offer higher research quality (Soderberg et al., 2021) and exhibit lower levels of publication bias (Scheel et al., 2021). Another crucial aspect of open science is providing access to materials, data, and code, which enables the replication and follow-up research to mirror study materials and procedures, and facilitates the reuse of data for secondary data analyses and research synthesis (e.g., in meta-analyses).

Although significant structural barriers, including journal editorial policies and a lack of incentives for individual researchers and their career needs, have prevented the full-scale adoption of open science practices,⁸ open science practices have the potential to enhance the scientific body of findings. These practices have gained increasing acceptance, especially in the health sciences, medicine, and psychology (Hagger, 2022; Nosek et al., 2018). Many funding agencies, including the National Institutes of Health (2023) and the National Science Foundation (2023), now encourage or mandate the adoption of open science practices. In addition, both the U.S. federal government and the European Commission have introduced open science initiatives (European Commission, 2023; The White House, 2023).

These developments have inspired a similar push for open science practices in management research. Discussions on implementing such practices have emerged in fields like marketing (Krishna, 2021; Pham & Oh, 2021; Simmons et al., 2021) and information systems (Bogert et al., 2021; Doyle, 2021). Leading journals, such as *MIS Quarterly* and *Nature Human Behavior*, are currently trialing registered reports and preregistration, along with the open sharing of materials.⁹ Focused journals, such as the *AIS Transactions on Replication Research*, offer a platform for publishing replication efforts, while journals like *Data in Brief* offer an outlet for sharing research datasets. Moreover, multiple journals now award open science badges to articles that feature preregistered studies, open data, or open materials (Center for Open Science, 2022). However, the extent to which open science practices have been adopted in many disciplines, including marketing research, remains unclear. In the following section, we discuss why open science practices can be especially helpful in PLS-SEM-based studies.

2.3. Motivation for adopting open science practices in PLS-SEM

In recent years, there has been a significant expansion in the methodological toolkit and application of PLS-SEM across management disciplines (Khan et al., 2019; Sarstedt, Hair, Pick, et al., 2022a). It is fair to say that PLS-SEM is one of the fastest-evolving statistical techniques, thanks to its numerous strengths and affordances (Sharma et al., 2023). Among its widely discussed strengths are its ability to balance explanatory and predictive goals (Liengaard et al., 2021; Richter et al., 2016; Sharma et al., 2023; Shmueli et al., 2016), conduct a simultaneous

⁷ <https://www.cos.io/initiatives/registered-reports>.

⁸ For more details on the practical and ideological issues associated with the adoption of open science practices, interested readers are encouraged to refer to Nosek et al. (2012), Nosek et al. (2018), and Cohoon and Howison (2021).

⁹ https://misq.umn.edu/call_for_papers/registered-reports; <https://www.nature.com/nathumbehav/submission-guidelines/registeredreports>.

⁶ <https://www.cos.io/initiatives/prereg>.

assessment of measurement and structural models (Petter & Hadavi, 2021), perform model comparisons (Danks et al., 2020; Sharma et al., 2019), assess complex models involving a large number of constructs, including higher-order and formative constructs (Cheah et al., 2019; Petter & Hadavi, 2021; Wetzels et al., 2009), carry out multigroup analysis (Cheah et al., 2023; Cheah et al., 2020; Sarstedt et al., 2011), and estimate conditional process models (Sarstedt et al., 2020). While the broad scope, flexibility, and ease of use of PLS-SEM make it suitable for various modeling tasks, these strengths also create opportunities for potential misapplication (Petter & Hadavi, 2021). For example, researchers have incorrectly justified the use of PLS-SEM in studies with unreasonably small sample sizes, based on its ability to converge and provide a solution (Marcoulides & Saunders, 2006; Sarstedt, Hair, Pick, et al., 2022a). Such underpowered studies are less likely to be independently replicable and raise concerns about publication bias (Etz & Vandekerckhove, 2016; Ioannidis, 2005).

Similarly, the strengths of PLS-SEM in prediction, flexible model specification, and handling complex models and data make it an attractive technique for exploratory theory development (Petter, 2018; Richter et al., 2016). The need for developing theories in rapidly changing or emerging domains requires the use of methods like PLS-SEM with a predictive and exploratory orientation to address phenomena that are difficult to explain using existing theories (Richter et al., 2016). While exploratory theory development has been popular in natural sciences such as meteorology and astronomy, the management sciences have traditionally relied on deductive or theory-confirmation approaches. However, recent calls to exploit the synergies between theory and data have increased interest in exploratory techniques in management disciplines for novel theory development (Berente et al., 2019; Danks et al., 2023; Maass et al., 2018; Shmueli, 2010). Initial theory development in such areas often follows a funnel-like approach, where researchers begin with generic questions driving the phenomena and engage in an iterative “dialog” between the emerging theory and the data to progressively narrow the focus of investigation (Sinkovics & Alfoldi, 2012; Wold, 1982). In fact, Wold (1982) envisioned PLS-SEM as a predictive “soft modeling” approach to discover new patterns in the data, where researchers estimate and compare multiple alternative models rather than confirming an *a priori* specified model (Chin, 1998; Richter et al., 2016; Sharma et al., 2019). When it is transparently conducted and reported, this method of exploration can lead to significant scientific advancements by enhancing the effectiveness and efficacy of theory development (Hollenbeck & Wright, 2017). However, researchers often do not disclose these exploratory assessments for various reasons, including journal requirements, reviewer preferences, and the pressures to publish articles with a clear and coherent narrative. Instead, they tend to selectively present their exploratory results as if they were derived from *a priori* hypotheses. In this regard, Petter and Hadavi (2021, p. 13) note that: “The balance between engaging in exploratory analysis and fishing for results can be tenuous. Much of this practice happens in the background and is not reported to reviewers, editors, and readers.” Open science practices such as preregistration and registered reports can bring this “exploration into the open” and help establish foundational theories in emerging domains.

Finally, the availability of various PLS-SEM software, metrics for structural and measurement models, bootstrap methods and samples, modes of measurement, options for higher-order constructs, missing data treatments, and moderation methods provides researchers with numerous degrees of analytical freedom. However, if research plans and materials are not made available for independent verification, it can lead to questionable research practices and false positive results (Simmons et al., 2011; Wicherts et al., 2016). The situation is further exacerbated by the rapid evolution of the PLS-SEM toolkit, which has rendered obsolete several model assessment and reporting practices that were acceptable not too long ago (Sarstedt, Hair, & Ringle, 2022b). Thus, while PLS-SEM is an invaluable technique with several analytical strengths, it is also susceptible to misapplications, much like any other

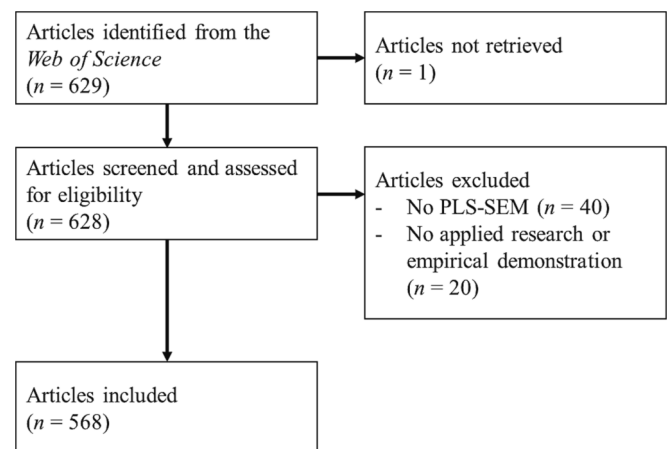


Fig. 1. PRISMA flowchart. Note: Fig. 1 shows the PRISMA flowchart outlining the article screening and selection resulting in 568 articles included in our analysis.

statistical technique (Petter, 2018; Wicherts et al., 2016). It is not surprising that Petter and Hadavi (2021, p. 10), in their message to PLS-SEM researchers, draw inspiration from the *Spider-Man* movie quotation: “with great power comes great responsibility.”

These critical issues, along with the broader scientific movement toward increased research transparency and forthcoming journal policy changes, offer strong motivation for adopting open science practices in PLS-SEM studies. Researchers must stay informed of these upcoming changes and not be caught off guard. However, researchers currently have an incomplete understanding of the extent of open science adoption in PLS-SEM research, and initial evidence suggests that open science practices are only beginning to emerge in research domains like marketing, where PLS-SEM is popular. To investigate this hypothesis, we conduct a structured literature review of marketing research to examine the prevalence of open science practices in PLS-SEM.¹⁰ We provide our materials, data, and analysis code on the OSF (<https://osf.io/46pyk>) and, in addition, include a similar analysis for journals in international business research—which is also a prominent area for PLS-SEM research. Based on these findings, we offer recommendations for how to transition PLS-SEM toward open science practices. Furthermore, considering the limited adoption of open science practices and the need for more replications, we advocate for the open sharing of study resources and propose a preregistration template tailored to PLS-SEM.

3. Structured literature review

3.1. Methodology

We conducted a structured literature review by following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) approach in our literature search and article screening procedure to assess the state of the art in open science and replication in PLS-SEM applications within the marketing domain (Moher et al., 2015; Page et al., 2021). We used the keywords “partial least squares” and “PLS” to conduct a full-text search in the *Clarivate Analytics Web of Science* for journal articles published from January 2012 to May 2022 in the top-tier (Q1) journals according to the 2021 SCImago Journal Ranking for the subject category Marketing (SCImago, n.d.). This database search yielded 629 articles (see Fig. 1). After one article’s full text could not be retrieved, we proceeded to screen the contents of the remaining 628

¹⁰ We focus on marketing research as it is a popular area for PLS-SEM applications and has been the subject of debates on open science practices and replicability in recent years (see, e.g., Babin et al., 2021; Krishna, 2021; Labroo et al., 2022; Pham & Oh, 2021; Simmons et al., 2021).

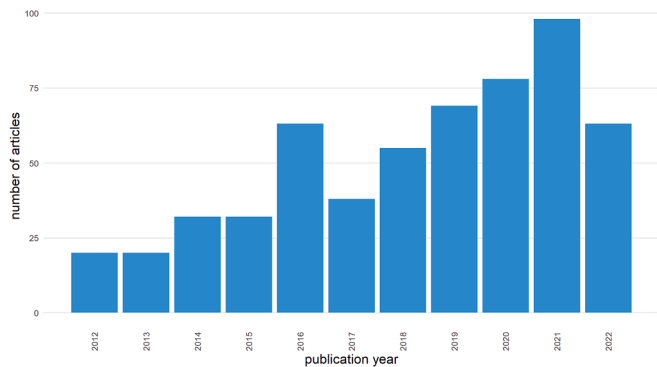


Fig. 2. Number of articles per year. Note: Fig. 2 shows the number of PLS-SEM articles in marketing research per year from January 2012 to May 2022 and depicts an increasing trend.

articles. Our review included only articles that used PLS-SEM in empirical or applied research or illustrated a methodological technique with an empirical example. Thus, we excluded review articles, purely methodological articles, and simulation studies from our analysis. This procedure resulted in a total of 568 articles for review.

We coded these 568 articles to identify open science practices and replication studies (see coding scheme in Table A.1 in the appendix). Within the selected articles, we identified a total of 586 individual studies that met our inclusion criteria.¹¹ For each such study within an article, we assessed whether it was preregistered, which platform and template were used for preregistration (e.g., OSF or aspredicted.org), and whether a valid link to the preregistration was provided. We also documented whether the article was a registered report and checked the data availability. Some authors may not provide datasets for privacy reasons or due to nondisclosure agreements. We coded whether a correlation or covariance matrix for latent variables or indicators was available and assessed whether study materials, including questionnaires, item wordings, stimuli, and other materials, were provided. Specifically, we checked whether the information provided for the measurement model (i.e., item wordings and allocation to constructs) was sufficient to reproduce the model. Our coding also captured the software used (e.g., SmartPLS, Ringle et al., 2022; or the *semr* R package, Ray et al., 2022) and, if applicable, whether analysis code was provided and whether the article had received an open science badge (Center for Open Science, 2022). Lastly, we identified replication studies and coded whether the replication was direct or conceptual; and referenced the original study.

3.2. Results

Among the 568 articles reviewed, 558 articles were applied research articles, and 10 were methodological articles that included an empirical demonstration meeting our criteria for analysis. As reported before (Khan et al., 2019; Sarstedt, Hair, Pick, et al., 2022a) and outlined in Fig. 2, PLS-SEM research has been growing in popularity over the past decade.

Further descriptive analyses revealed that 24 journals in our dataset published applied PLS-SEM research during the 10-year period, with the *Journal of Business Research* ($n = 164$ articles with 171 studies) and the *Journal of Retailing and Consumer Services* ($n = 135$ articles with 138 studies) as the most receptive journals. Overall, our findings indicate a low prevalence of open science practices in all but one of the major coded categories (Fig. 3).

Specifically, our dataset includes only one preregistered study

(Nunes et al., 2021) and no registered reports. None of the journals in our dataset have issued open science badges. Data is available for only five studies, of which three are empirical demonstrations in methodological articles (data stored on pls-sem.net) and two are applied research (data stored on Mendeley Data).¹² We identified only four replication studies, with three of them being within-article replications. This share of replications is lower than comparable metrics reported in other studies. For example, Ryan and Tipu (2022) report a replication article share of 3.0% in a hand-coded sample of 500 articles randomly selected from top-tier marketing journals indexed as 4* & 4 in the 2018 Academic Journal Guide. Among the studies that indicate the use of software ($n = 480$, 81.9%), SmartPLS is the most commonly used software ($n = 432$), but only one study provides a SmartPLS archive file (Graciola et al., 2018). Furthermore, while eight studies use R for their analyses (see Fig. 3 for package information), only one study provides the R code (Hult et al., 2018).

While the aforementioned results indicate a relatively low prevalence of open science and replication practices within PLS-SEM in marketing, most studies offer materials such as scale items ($n = 512$; 87.4%). However, these materials are exclusively provided within the article itself or the article's supplementary materials. We identified nine studies with a full questionnaire available, 478 studies in which scale items are fully provided, 30 studies in which scale items are only partially provided, and 34 studies with other available materials (e.g., stimuli).

Moreover, we conducted a more detailed assessment of material availability, including the availability of a full description of survey items (i.e., the wording of the survey items), and found that it varies between publication years and journals (Fig. 4). Specifically, we found that the availability of item wording ranges from 65.5% (2016) to 90.2% (2020) across different years, but we did not observe a clear longitudinal trend (upper part in Fig. 4).

Limiting our analysis to journals with at least 10 studies, we found that the full description of survey items is highest in *Industrial Marketing Management* (100.00%), the *Journal of Marketing Management* (95.7%), and the *Journal of Services Marketing* (95.5%). Conversely, the journals that had the lowest percentages were *Psychology & Marketing* (73.3%), the *Journal of Business Research* (69.01%), and the *International Business Review* (64.3%).

Finally, we examined the potential relationship between following open science practices or conducting replications and the number of citations an article receives. We found that following at least one open science practice or replicating another study within an article ($n = 516$ articles; article citation: $M = 32.4$, $SD = 54.7$, $median = 15$) versus none of these ($n = 70$ articles; article citation: $M = 36.1$, $SD = 53.5$, $median = 21.5$) is not linked to the article's number of citations (Wilcoxon test: $W = 20,092$, $p = 0.126$; t -test with logarithm used for citations: $t(89.52) = 1.31$, $p = 0.194$, Cohen's $d = 0.16$). A Bayesian t -test using logged citations as a dependent variable provided moderate evidence in support of the null hypothesis ($BF_{01} = 4.46$). The relationship between citations and open science practices or replication remained unchanged when controlling for other factors, such as the journal's impact factor (Web of Science Group, 2022), the article's publication year (linear and quadratic trend), and whether the article is available for open access (see Table 1 for details). Importantly, we find a positive association between open access and citations, indicating that the free availability of the article increases its reach and impact ($b = 0.29$, $SE = 0.09$, $t = 3.25$, $p = 0.001$).

¹¹ The dataset includes 551 articles with one study in our analysis, 16 articles with two studies, and one article with three studies.

¹² data.mendeley.com.

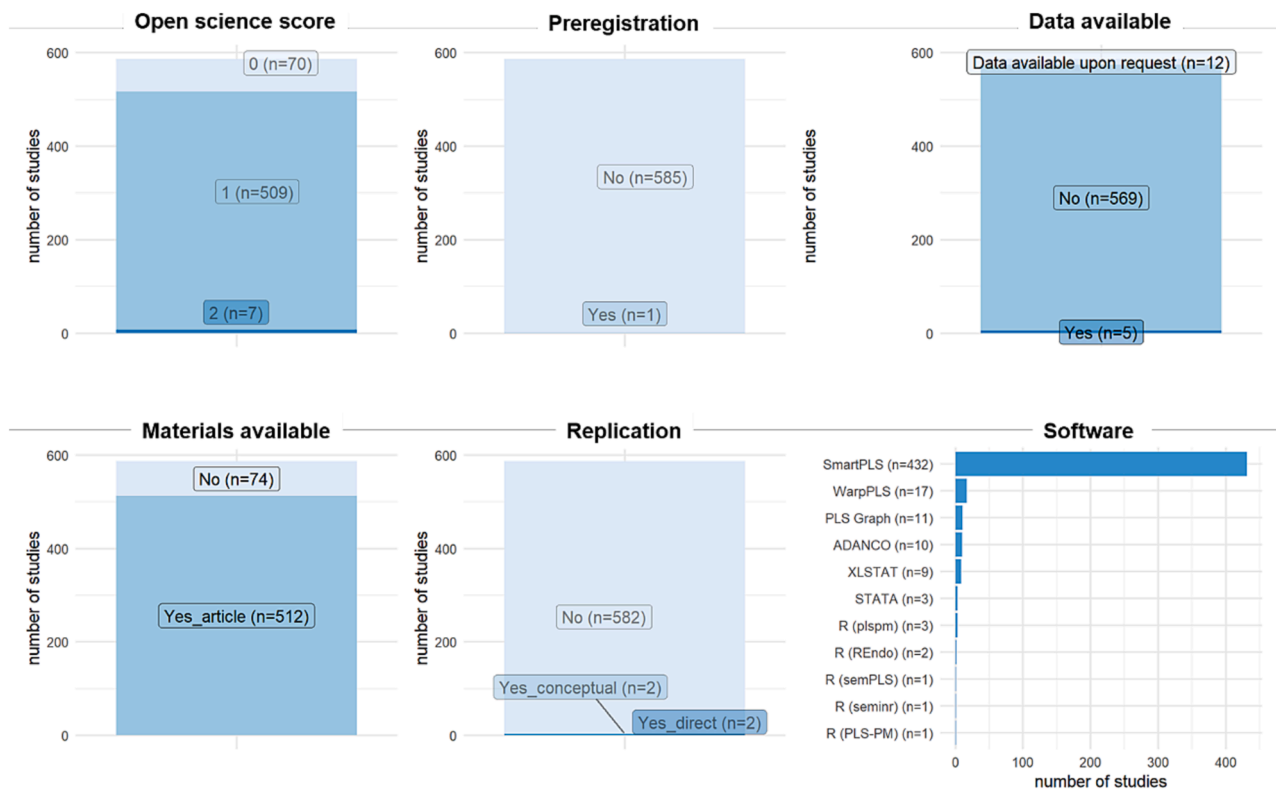


Fig. 3. Prevalence of different open science practices, replication, and software used. *Note:* Fig. 3 shows the number of studies that employed open science and replication practices as well as applied software (lower right panel). The open science score (upper left panel) is the number of employed open science and replication practices (pre-registered, data available, materials available, analysis code available, replication; all coded as yes[1] and no[0]). The open science score, therefore, has a maximum of 5 and a minimum of 0.

4. Discussion

4.1. Moving PLS-SEM research toward open science

Situated at the intersection of the debates surrounding the adoption of open science practices in general and the misapplication and misreporting of results in PLS-SEM in particular, our in-depth analysis of PLS-SEM studies in leading marketing journals reveals that while most researchers offer adequate information to merely reproduce their model graphs (i.e., they only share the relevant item wordings and their relationships in the model), they do not share all materials that facilitate independent efforts to reproduce or replicate their findings.

We find an overall low usage of open science practices (e.g., pre-registration, open data, open material, open code, or open model files) in the field of PLS-SEM. This inadequate use of the practices hinders progress as the information necessary for building on or replicating prior research remains classified. Such a status quo is incompatible with the principles of error-correcting science. Unfortunately, this opaqueness reinforces criticisms regarding the unwarranted utilization of PLS-SEM's strengths, particularly its analytical flexibility, which, when appropriately and transparently employed, can lead to major research advances (Richter et al., 2016). We believe that encouraging open science practices in PLS-SEM can provide a way to break the status quo. To this end, we offer two recommendations: first, to adopt preregistration, which entails rigorous research planning and the disclosure of study and analysis plans prior to data collection (or before data analysis in the case of secondary data); and second, to provide materials and data that are reusable for further research whenever possible, thereby ensuring the reproducibility and replicability of PLS-SEM results.

To enhance rigorous research planning, we recommend preregistering PLS-SEM research analyses. This practice helps disentangle planned analyses and post hoc exploratory analyses and transparently

report any analytical degrees of freedom utilized in the analyses. To facilitate the process of preregistration, we propose a PLS-SEM template that researchers can use to provide more insight into their analyses.

At this point, readers may wonder why a specific template is needed when several preregistration templates are already available (e.g., Bowman et al., 2020; van 't Veer & Giner-Sorolla, 2016; Wharton Credibility Lab, 2023). We offer three reasons: First, the available preregistration templates primarily focus on study design aspects (e.g., the goal of the study, hypotheses, sample, and data curation) but not on the specific methodological choices researchers often make during statistical analysis.¹³ These choices, however, have a known impact on statistical results and can contribute to publication bias (Silberzahn et al., 2018; Wicherts et al., 2016). Second, PLS-SEM applications involve complex data and models that require distinct goals, model assessment practices, and reporting requirements compared with other multivariate techniques, like covariance-based SEM (Hair et al., 2017). Third, researchers may utilize the PLS-SEM toolkit for various purposes, including developing theory, focusing on explanation, prediction, or both explanation and prediction, or using one-dimensional versus multidimensional constructs (Petter & Hadavi, 2021). Each of these goals, however, requires different sets of metrics and tests within the PLS-SEM framework to be reported. By introducing a PLS-SEM-specific preregistration template, we provide researchers with a structured approach to guide their study design and streamline their efforts. In proposing this template, we are mindful that it is not intended to be a strict "one-size-fits-all" template. Instead, it serves as a facilitating

¹³ We are aware of a methodology-oriented preregistration template that offers specific reporting requirements tailored to psychometric network analysis (Burger et al., 2022) and another tailored to secondary data analysis (van den Akker et al., 2021).



Fig. 4. Share of studies providing materials (materials provided) and all item wordings associated with the model (all items available). *Note:* Sample size (n ; indicated on y-axis labels) corresponds to the number of studies per year, respectively per journal. The x-axis shows the share of studies and is scaled to 100% corresponding to the number of articles per year/journal. Dot sizes correspond to the number of studies per year/journal. Dot colors indicate whether any materials were provided (grey dots) and whether all items were available (blue dots). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

“nudge” to researchers who may use it to enhance transparency as much as possible. We recommend utilizing this template whenever feasible.

We derived the template by using state-of-the-art literature (e.g., Becker et al., 2023; Hair et al., 2022; Hair et al., 2018) and engaging in discussions with two PLS-SEM experts. Subsequently, we asked four more researchers with extensive PLS-SEM expertise to pretest the pre-registration template. The proposed PLS-SEM preregistration template comprises five sections (see Table A.2 in the appendix).¹⁴ The first section (“general research setup”) focuses on the research aim and procedure and precedes the second section on the “data preparation and analysis plan,” which considers specific analytical settings, choices, and procedures. The remaining three sections allow researchers to add information on planned robustness checks, potential exploratory analyses, and software they plan to use. It also prompts them to make concluding statements. The template shows some suggested answers for standardized workflows (e.g., common procedures for measurement model evaluation). In addition, we provide a document on the OSF with supplementary information on how to use the template, how to report

deviations from the preregistered plan, and suggested readings on the outlined concepts. Furthermore, we have included an example template with the model provided by Hair et al. (2022). We recommend submitting the template to the OSF registries under the open-ended OSF-preregistration template, which is a pre-defined, free-format template that allows appending the completed PLS-SEM preregistration template and additional files, such as questionnaires or analysis codes.¹⁵ To further enhance reproducibility, we encourage open sharing of materials, data, and analysis code, whenever possible, to enable reuse in further research, replications, or meta-analyses. Although our analysis in the marketing domain suggests that open-access PLS-SEM articles receive more citations than non-open-access articles, we do not observe a similar pattern in our coding of open science or replications. Merely sharing item wordings may not attract citations, since researchers are more likely to cite the original study that developed those scales. However, a study’s impact may increase when the full set of materials (including data) is made available.

Online repositories, such as OSF, provide an open-access and flexible

¹⁴ We provide a ready-to-use template on the OSF: <https://osf.io/46pyk>.

¹⁵ <https://osf.io/registries>.

Table 1

Regression on article citations (logarithm) as the dependent variable with open science practices and replication, as well as other covariates.

Predictors	Estimates (beta)	SE	t	p	CI (95%)
(Intercept)	-228,197.87	19,649.08	-11.61	<0.001	[-266,789.89, -189,605.86]
Open science or replication*	0.15	0.12	1.24	0.214	[-0.08, 0.37]
Impact factor	0.07	0.010	6.93	<0.001	[0.05, 0.09]
Publication year	226.52	19.48	11.63	<0.001	[188.26, 264.78]
Publication year ²	-0.06	0.005	-11.65	<0.001	[-0.07, -0.05]
Open access	0.29	0.090	3.25	0.001	[0.11, 0.47]
Observations	586				
R ² / R ² adjusted	0.555/ 0.551				

Note: * Open science or replication is the dichotomized open science score (see Fig. 3) with 0 = neither open science practice nor replication and 1 = at least one open science practice or replication.

space to store research materials. In addition, OSF offers a platform for sharing supplementary analyses (e.g., deviations from the original plan, more exhaustive data descriptions, and visualizations) that are too extensive to be placed in the published article and would otherwise remain unpublished. PLS-SEM software typically provides convenient means for sharing data, models, and results. For example, SmartPLS allows the export of model folders, including data and the model setup, and facilitates the online sharing of result reports (Ringle et al., 2022). Researchers can also share code from statistical languages such as R (R Core Team, 2023) and provide result reports in formats like Rmarkdown or quarto documents (Allaire, 2022; Allaire et al., 2022).

While researchers often have the voluntary choice of engaging in preregistration and material sharing, other approaches require the involvement of publishers and editors. In response to the growing emphasis on open science, several journals now encourage preregistration and data sharing (Labroo et al., 2022) or offer registered reports (Nosek & Lakens, 2014).¹⁶ Establishing visible incentives, such as awarding open science badges that signal open data or open materials, as well as preregistration, also helps foster positive attitudes toward data and material sharing (Kidwell et al., 2016). Finally, there are calls for institutionalizing methodological review boards to evaluate planned studies before data collection (Lakens, 2023). As such, the proposed PLS-SEM preregistration template can be a valuable tool for researchers.

4.2. Limitations

We will now discuss some limitations of our work. First, given the apparent scarcity of research at the intersection of PLS-SEM and open science, we decided to conduct an in-depth examination of current research practices and focused our structured literature review on marketing journals as a prominent domain for applied PLS-SEM research. While our analysis reported in the main paper offers insights specific to the field of marketing and our additional analysis reported on the OSF focuses on international business research, existing literature suggests that a similar lack of adherence to open science practices and replications exists in other management fields (Cohoon & Howison, 2021; Ryan & Tipu, 2022; Tipu & Ryan, 2022). Thus, our findings reflect a broader pattern across various scientific disciplines.

Second, our emphasis on recommending open science practices in PLS-SEM studies should not be misconstrued as a critique of the method itself. The issues we have discussed broadly affect all statistical methods (Simmons et al., 2011; Wicherts et al., 2016). However, given the scrutiny surrounding PLS-SEM applications, we believe that open science offers a practical approach to addressing the aforementioned problems by promoting transparency in the planning, execution, reporting, review, publication, and verification of research. This can

help increase confidence in PLS-SEM studies.

Third, recommending a preregistration template may initially seem like an additional burden for researchers. However, we believe the opposite to be the case. Providing concrete study objectives, procedures, and data analysis plans helps researchers better prepare and anticipate potential methodological issues in advance (e.g., sample size requirements). Thus, adhering to a reasoned original data analysis plan can save researchers time by avoiding the pitfalls of exploring multiple analysis paths, commonly referred to as the “garden of forking paths” in analytical research (Gelman & Loken, 2014). Furthermore, the preregistration template serves as a valuable guiding document for researchers to conduct more robust analyses in line with the latest empirical guidelines.

Fourth, researchers may have concerns that preregistration could restrict their analytical freedom to conduct exploratory data analyses. As we pointed out before, researchers can still conduct post hoc exploratory analyses and report these results. Preregistration can encourage open communication and transparent reporting of exploratory analyses, thereby enhancing the credibility of research that employs both confirmatory and exploratory approaches. Preregistration is also flexible and allows authors to communicate any deviations from the original research plan. Deviations from an initial study plan are inevitable in any research endeavor, as they allow for the correction of potential errors in the study or the adoption of better approaches as knowledge accumulates throughout the research process (Nosek et al., 2019).

Finally, it is important to note that preregistration does not prevent researchers from utilizing a dataset in multiple studies. Researchers can issue separate preregistrations for subsequent studies by providing transparent plans for each one. However, we caution against the practice of *preregistration hacking*, where researchers issue a series of preregistrations for the same study after having viewed the results of the analyses. Platforms like OSF offer time-stamped indexing to keep a record of all preregistrations issued, thereby deterring such hacking. The key takeaway is that open science practices do not restrict researchers; rather, they enable researchers to communicate their findings more confidently.

5. Conclusion

Our review points to a lack of adherence to open science practices in PLS-SEM-based research, which poses the risk of perpetuating false positive findings. Given ongoing discussions surrounding the misuse and misreporting of results and the calls to improve methodological rigor (Petter & Hadavi, 2021; Sarstedt, Hair, & Ringle, 2022b), we believe that embracing open science practices can help alleviate these concerns by fostering transparency in PLS-SEM-based studies. We advocate for the open-access sharing of resources in PLS-SEM research. Our proposed PLS-SEM-specific preregistration template represents a step in this direction and can be used to improve research transparency, thereby enhancing confidence in PLS-SEM-based studies.

¹⁶ You can find a list of journals offering registered reports at <https://www.cos.io/initiatives/registered-reports>. Within our structured literature review, *Psychology and Marketing* is at the time of writing this article the only journal that offers registered reports.

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CRediT authorship contribution statement

Susanne Jana Adler: Conceptualization, Project administration, Supervision, Formal analysis, Visualization, Writing – original draft preparation, Writing – review & editing. **Pratyush Nidhi Sharma:** Conceptualization, Writing – original draft preparation, Writing – review & editing. **Lăcrămioara Radomir:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared further materials, such as data and code, on the OSF.

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Appendix

Open science coding scheme

(See [Table A1](#)).

Table A1
Coding scheme.

Category	Specific coding
Pre-registration	Was the study preregistered? <i>Yes/No</i> If a study was preregistered, indicate the platform (e.g., OSF, aspredicted) If a study was preregistered, indicate the template. (List of different templates available at the OSF: https://osf.io/zab38/) Can the preregistration be accessed/viewed? <i>Yes/No</i> Copy/paste the preregistration link
Registered report	Was the study a registered report? <i>Yes/No</i>
Open/ accessible data	Is the data available? <i>Yes/ Data available upon request/ No</i> Indicate data storage platform (e.g., OSF) Copy/paste the link to the data Indicate whether the data is available from the platform. <i>Yes/No</i> Indicate whether correlation/covariance matrices are available with coding. - <i>Available for latent variables/constructs</i> - <i>Available for indicators</i> - <i>Available for both latent variables and indicators</i>
Open/ accessible materials	Indicate whether any materials for the study (stimuli, questionnaire, etc.) are available in an open repository (e.g., OSF) or stored with/in the article If materials are stored at an open repository (e.g., OSF) - Indicate the platform - Copy & paste the link to the materials - Is the complete material (incl. all elements such as questionnaire, stimuli, etc.) available on the platform? <i>Yes/No</i> - Is the questionnaire available on the platform? <i>Yes, complete/ Yes, partial/ No</i> - Are the items/scales available on the platform? <i>Yes, complete/ Yes, partial/ No</i> - Are the stimuli available on the platform? <i>Yes, complete/ Yes, partial/ No</i> - Specify other materials available on the platform. If materials are provided with/in the article - Is the complete material (incl. all elements such as questionnaire, stimuli, etc.) available in the article? - Is the questionnaire available in the article? <i>Yes, complete/ Yes, partial/ No</i> - Are the items/scales available in the article? <i>Yes, complete/ Yes, partial/ No</i> - Are the stimuli available in the article? <i>Yes, complete/ Yes, partial/ No</i> - Specify other materials available in the article. Do the authors provide item wordings (<i>items provided</i>) or references to the scales used (<i>references provided</i>)? - Other possible coding: <i>mixed</i> (i.e., sometimes items provided, sometimes references provided) - If necessary, specify other categories for item sharing. Are all indicators associated with the constructs reported? <i>Yes/No</i>
Software	Which software was used for the analysis? If applicable, indicate the package (e.g., for R)
Open/ accessible code	Is the data analysis code (e.g., R code) available? <i>Yes/No</i>
Open science badge	Does the article have an open science badge? <i>Yes/No</i> Which open science badge (if applicable)?
Replication	Do the authors replicate a prior study and claim to do so (i.e., explicitly label their study as a replication)? <i>Yes, direct/ Yes, conceptual/ No</i> Refer to the original study (e.g., link or doi)

Note. Please see OSF supplementary material for more details, such as specific variable names (<https://osf.io/46pyk>).

PLS-SEM preregistration template questions

(See Table A2).

Table A2

Sections and questions in the PLS-SEM preregistration template.

1. General research setup	
1.1	Research question and hypotheses <i>Please describe the research question(s).</i>
1.2	Hypotheses <i>Please describe the hypotheses.</i>
1.3	Structural model <i>Please provide a clear description of your structural model and include the corresponding model graph below.</i>
1.4	Measurement model <i>Please describe the measurement model. For each construct, include the details of how you will measure its indicators (e.g., item wording, scale), the number of items per construct, and its mode (e.g., reflective, formative). For this purpose, you can use the two tables below.</i>
1.5	Rationale for using the analysis technique <i>Please specify the method(s) employed in this research (e.g., PLS-SEM and any other methods). Please provide the rationale for using PLS-SEM in this research. Please outline whether your study focuses on prediction only, explanation only, or both, and briefly explain your rationale.</i>
1.6	Study procedure <i>Please describe the study design and the study procedure. If applicable, list any experimental factors or other data characteristics induced by the study design. List the measures, manipulations, and study materials used, as well as their order of presentation to the participants.</i>
1.7	Sample size <i>Please specify the planned sample size and how you determined it, indicating the method used (e.g., inverse square root method, power analysis).</i>
1.8	Data collection
1.8.1	Primary data <i>If you are using primary data, please describe the data collection procedure (e.g., lab study, online survey, MTurk sample). If applicable, describe the characteristics of the target sample or any pre-selection rules (e.g., representative sample for the population).</i>
1.8.2	Secondary data <i>If you are using secondary data, please indicate the data source and describe any previous publications (e.g., research publications or data description publications) where the secondary data has been used, as well as how your research relates to these publications. If applicable, describe any pre-selection rules (e.g., drawing a random sample of the data or selecting a subgroup).</i>
1.9	Data availability <i>Please describe your plan to make the data available, if applicable. If you analyze secondary data, please indicate whether the data is publicly available.</i>
1.10	Common method variance <i>Please select if common method variance (CMV) may affect your results. If applicable, please select how you plan to assess CMV. If applicable, please describe your CMV assessment.</i>
1.11	Endogeneity <i>Please select if endogeneity may affect your results. If applicable, please select how you plan to assess endogeneity. If applicable, please describe your endogeneity assessment.</i>
1.12	Additional information <i>If there is any additional information regarding the research setup, data, or data quality checks that you would like to preregister, please provide it here.</i>
2. Data preparation and analysis plan	
2.1	Data cleaning <i>Please describe the planned data cleaning procedure and exclusion criteria (e.g., removal of participants who failed attention checks, removal of outliers, removal of straight-liners).</i>
2.2	Missing value handling <i>Please select how missing values will be handled. If applicable, please specify the procedure.</i>
2.3	Path model estimation <i>Please specify the algorithm used for the path model estimation (e.g., standard algorithm or consistent PLS (PLSc)) and its specific settings (e.g., path or factor weighting scheme). Additionally, describe the settings for the bootstrapping procedure used to determine the statistical significance of the results (e.g., number of bootstrap subsamples and confidence interval computation method). If applicable, describe any additional settings, such as the creation of interaction terms (e.g., two-stage approach). If applicable, describe a data weighting procedure.</i>
2.4	Assessing the measurement model
2.4.1	Reflective measurement models <i>Please describe the evaluation of the reflective measurement model. Specifically, specify how you will assess measurement reliability and validity, which threshold values you will apply, and the course of action if those threshold values are not met (e.g., if and in which situation indicator removal is considered)</i>
2.4.2	Formative measurement models <i>Please describe the steps in the formative measurement model evaluation. Specifically, please specify how you will assess measurement convergent validity and collinearity between indicators, as well as the significance and relevance of outer weights. In addition, indicate the threshold values you will apply and describe how will proceed if those threshold values are not met (e.g., if and in which situation indicator removal is considered).</i>
2.5	Assessing the structural model <i>Please describe the assessment of the structural measurement model.</i>
3. Robustness checks and additional analyses	
<i>Please describe any robustness checks and additional (complementary) analyses you plan to conduct (e.g., importance-performance map analysis, multigroup analyses, necessary condition analysis, latent class analysis). If applicable, please specify if you plan to conduct exploratory analyses and briefly describe their aim (e.g., if you plan to make model adjustments that could result in an exploratory model comparison analysis).</i>	
4. Software	
<i>Please list the software (if applicable, the specific packages) you plan to use for your analysis. If applicable, please provide a code draft (e.g., an R script) together with the preregistration.</i>	
5. Final statements	
5.1	Data collection statement <i>Has data collection already begun for this study?</i>
5.2	Additional information <i>Any additional information or comments you would like to preregister?</i>

Note. The template follows the model evaluation recommendations as described in Hair et al. (2022). We provide multiple template documents on the OSF (<https://osf.io/46pyk>).

We provide a word document and form as well as a document with a list of all sections, instructions, options, and additional information.

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